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Vet Record SUPPLEMENT

SURGICAL CASTRATION OF DOGS

Are there grounds to rethink?

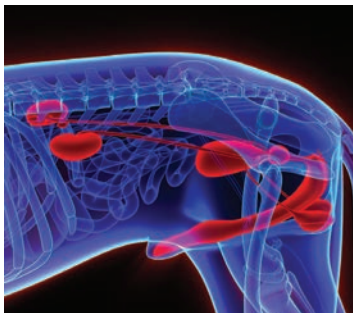
BVA British Veterinary Association

In association with Virbac

CONTENTS

1 TO CASTRATE OR NOT TO CASTRATE? THAT IS THE QUESTION...

3 'NEUTERING IS NO LONGER THE HALLMARK OF RESPONSIBLE PET OWNERSHIP'



5 BEHAVIOURAL ISSUES: CASTRATION MAY NOT BE THE SOLUTION

7 CHARITIES CONSIDER THEIR CASTRATION POLICIES

9 AN OWNER'S PERSPECTIVE

FOREWORD

Time for new conversations about canine castration

IN the UK, the idea that neutering is a key component of responsible animal ownership has been long established. Canine castration has often been viewed alongside the likes of vaccination and parasite control: an action to be taken early in an animal's life to protect its health and that of the population more generally.

This has led to the UK having one of the highest rates of dog castration in Europe. Yet a growing body of research suggests that gonadectomy is associated with an increased risk of certain diseases, including cancer and joint conditions. It can also have a negative impact on an animal's behaviour.

In light of this, the positions of various veterinary organisations on reproduction control are evolving. The World Small Animal Veterinary Association is working on new guidelines in this area, and in the past few years in the UK, Dogs Trust and the RSPCA have both refined their approaches.

There are reports that owners' attitudes are evolving too. Vets and veterinary nurses report that more pet owners are asking questions about surgical castration, and are keen to understand the potential impact of the procedure on their animals.

This supplement, which will be followed by a roundtable event and a separate panel session at London Vet Show, is designed to review some of the key issues around surgical castration of dogs. Why has it been seen as best practice in the UK for so long? What is the evidence of the impact of castration on the overall and behavioural health of an animal? And what does this evidence mean in a practical sense for how individual veterinary teams should approach the issue of reproduction control in animals?

As veterinary professionals, our job is to do what is best for the animals under our care. When doubt arises over what we had long thought was best practice, it can be challenging. It's particularly difficult in the context of the many pressures under which the profession is operating. General practitioners have severely limited time and are, by their very nature, generalists. Following in-depth research across every sphere of veterinary practice is not realistic.

So we hope this supplement, prepared independently by the *Vet Record* editorial team and based on interviews with a range of professionals, serves to helpfully summarise the key considerations around castration. Its ultimate conclusion is that more conversations are needed about the practice, both within the profession and with owners. This supplement aims to help start some of those conversations.

Neil Mottram
Technical manager, Virbac UK

Virbac will be sponsoring one of the BVA Congress sessions at the London Vet Show. 'Is neutering dogs doing more harm than good?' will take place on Thursday 16 November 2023 at 10:30.

***Vet Record* has also run a roundtable to explore the issues around castration. A report from this event will appear in the 2/9 December 2023 issue.**

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To castrate or not to castrate? That is the question...

There is growing evidence that surgical castration of dogs may not bring all the benefits previously assumed. Times have changed too, and vets and clients are increasingly questioning whether routine castration should continue to be considered best practice.

ASK Benedicte Helljesen how many surgical castrations she performs on dogs each year and she has to think – not because there have been so many, but because there have been so few.

'None last year,' she says. 'Or maybe one? So far this year I think I've had two.'

Hearing this statistic, the assumption might be that Helljesen is a highly specialist vet. But the lack of frequency with which she performs castration is simply a reflection of the nation in which she practises – Norway, where neutering of dogs for anything other than an established medical need has been outlawed since 1974.

That legislative position is grounded in a belief that surgical procedures should only be performed for the welfare of an animal, not to adapt that animal to the needs of people. Accordingly, surgical castration is not seen as a legitimate answer to the question of population control but as an unethical and unnecessary intervention. ('When it comes to dogs, I think that it's the owners' responsibility to keep an eye on them, and keep them on leash,' says Helljesen.)

But the legislation also speaks to increasing concerns about the possible impact of surgical castration on an animal's health.

'I think most vets in Norway agree that we should be quite strict with this, because there are a lot of disadvantages of surgical castration,' says Helljesen, who is a veterinary surgeon at A-Vet in Larvik, south-east Norway.

'Castration is quite a quick surgery, but what you remove is quite valuable for the animal. It is removing something that has health benefits for the animal to keep.'

This is a contention increasingly backed by research. 'One of the things I often tell people is that we need to rethink the purpose of gonads, not just in ourselves as humans but in pets,' says Michelle Kutzler, professor of theriogenology at Oregon State University.

'They aren't merely reproductive organs,' she continues. 'They're also critically important for endocrine health, immune health, musculoskeletal health, antineoplastic health and behavioural health. Removing the gonads has negative impacts on many systems unrelated to the reproductive system.

'The complications, the side effects, the adverse effects if you will, of removing the reproductive tract have been well documented.'

Yet the strict approach to surgical castration seen in Norway stands in contrast to that of other nations – the UK included. Almost 70 per cent of dog owners surveyed for the 2023 PDSA Animal Wellbeing (PAW) report said that their animal had been neutered – one of the highest rates in Europe.

'In the UK, yes, we do a fair amount of neutering and it's traditionally been recommended for a whole variety of reasons,' says Anna Judson, who spent 30 years as a general practice vet and became president of the BVA in September.

In part castration was viewed as a means of reducing undesirable behaviour considered primarily hormone-driven – the likes of mounting and humping, urine marking and roaming to find bitches in season. It has also been seen as a means of preventing testicular cancers and potentially reducing the risk of some prostate and hormone-related problems. But, more broadly, there has been a strong population control argument.

> **Castration is quite a quick surgery, but what you remove is quite valuable for the animal**

'Over the initial years when I was in practice, it was definitely recommended for preventing unwanted pregnancies and population control,' says Judson. 'Traditionally in multipet households, where you've got male and female dogs, people would want to neuter to ensure there were no unwanted pregnancies from that.'

Neutering as best practice was very much 'a reaction to the stray situation as it was back then', suggests Caroline Allen, chief veterinary officer of the RSPCA. 'It was a reaction to litters of unwanted puppies being dumped and drowned, large numbers of stray dogs, and shelters having to euthanase healthy animals because there were just too many. There was a real need for that broader population control.'

It means that castrating a dog has become synonymous with responsible pet ownership. But for Kutzler, the reverse is actually true.

'It is a myth that to control the dog population you have to amputate the dog's gonads,' she argues. 'If you are a responsible pet owner and keep your dog on a leash, away from other intact dogs, that would control the population. That is the hallmark of responsible pet ownership.

'If you just open your door and let your dog run amok, not only could it breed other dogs but it could get hit by a car, or be stolen, or attacked by another animal. That to me is the hallmark of an irresponsible owner.

'If you are an irresponsible dog owner, neutering is probably the best option for you. But if you're a responsible dog owner, you should be considering other options.'

This is far from a minority view among theriogenologists, but it has taken time for it to gain traction more widely.

Stefano Romagnoli is professor of small animal reproduction at the University of Padova and president of the Italian Society for Animal Reproduction. In 2017, he suggested the creation of a committee on reproduction control within the World Small Animal Veterinary Association. Four years later, in 2021, it was finally established. The committee is currently working on new guidance about animal reproduction.

For Romagnoli, who chairs the committee, 'spaying and neutering should not any longer be considered best practice per se'.

'That doesn't mean it's necessarily worst practice,' he says. 'We can continue spaying and neutering, but I think it must be with a much higher degree of consciousness of what we're doing.'

Effectively, that means veterinary teams having more nuanced conversations with individual owners about surgical castration: exploring what the procedure would mean for that individual owner and that individual animal, in their specific circumstances. Do potential harms outweigh any potential benefits?

'The profession as a whole is talking about contextualised care and I would say that neutering is one of those decisions that needs to be contextualised to a particular animal – its breed, its age, its temperament, its circumstances, the wider pressures on the household, particularly in the cost-of-living crisis,' says Judson.

> **Spaying and neutering should no longer be considered best practice per se**

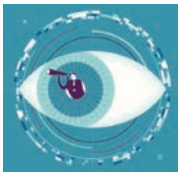
She recognises that sort of care takes more time, and suggests that organisations such as the BVA have an important role to play in providing easy means of decision support. 'It's almost going through a flowchart with owners to identify what the animal welfare priorities are, what the owners' priorities are, what the circumstances are in which that animal is living – are there both male and female dogs in the household?'

According to Steph Worsley, it's a conversation that increasing numbers of owners are keen to have. Worsley, a member of the British Veterinary Nursing Association council, has been a practising veterinary nurse for 24 years and says discussions about castration have evolved over that time.

'Twenty-four years ago, there were no ifs, buts or maybes – you recommended castration, and the sooner the better. At the practice I'm in, we're now taking each dog on a case-by-case basis, assessing whether castration would be a suitable option for them.

'Male owners, I have to say, have always been a little more hesitant about castrating their dogs – for obvious reasons, I think,' she says with a laugh. 'But more and more clients are definitely starting to think: "Do we need to do it?" If they've got good control of their dog, they know their dog is not going to be jumping six-foot fences when they smell a bitch in heat, then I think they're asking themselves: why should they castrate?'

EXPERT VIEWPOINT



'If surgical neutering is not the right option for every dog in every circumstance, then the question becomes what else can we do when birth control is needed for the welfare of that animal or for the family to function. We need a wide range of medical options – a breadth of options for the breadth of conversations we're now having.'

Anna Judson, *president, BVA*

'Neutering is no longer the hallmark of responsible pet ownership'

For some veterinary reproduction experts, the idea that responsible dog ownership and neutering are synonymous now needs to urgently change.

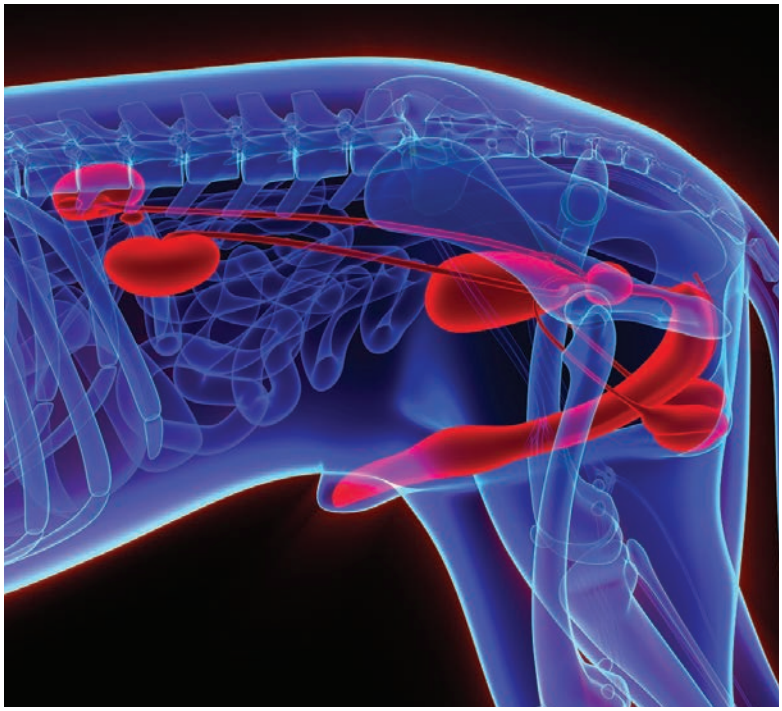
AS professor of theriogenology at Oregon State University, Michelle Kutzler says one of her main aims is to share her 'wonder, amazement and fascination' with dog reproduction. At the same time, she realises that the information she can impart – and, more specifically, the doubts it can introduce about best practice in animal birth control – may not be immediately welcome.

'I understand that this could be really frustrating for other veterinarians who, like me, had five lectures on dog reproduction in their entire studies and were pretty much told: "Neuter everything and then you don't have to worry about it." So I try to come from that lens: it might be confusing, but don't let that be a hurdle. Come with me on this journey and I'll explain what we know and we'll really understand it together.'

She is especially keen for vets to understand the breadth of ways in which research has shown the reproductive system contributes to an animal's physiology. Take for instance the role of luteinising hormone (LH), which is responsible for triggering ovulation in females and production of testosterone in males – and is a key research focus for Kutzler and colleagues at her lab.

'When we first started looking at it, we found luteinising hormone had receptors on lymphocytes, so these immune cells that are floating around and involved in everything from fighting off viruses, fighting off bacteria, to making antibodies. When we started to review the literature, we found that other people before us had demonstrated there were these receptors in humans, laboratory animals and even in dogs.'

In itself, that didn't necessarily mean a great deal. 'Just because a cell has a receptor for a certain hormone doesn't mean that it actually does anything. If you think of a receptor as a lock and the hormone as the key, you could unlock the door and there's nothing behind it,' she explains.



Indeed, Kutzler suspects the general assumption had been that there was nothing behind the LH receptor lock on the lymphocyte door. 'Not that there was any science behind it, but it was just: "Why would there be reproductive hormone receptors on lymphocytes? They must not be functional."'

To test the assumption, Kutzler and colleagues collected lymphocytes from both neutered and intact dogs. 'We grew those lymphocytes in cultures and added increasing levels of LH to see what happened. Did it unlock the door and there was something behind there? Sure enough, what happens is LH induces neoplastic changes in lymphocytes. Among them are cell proliferation, cell adhesion and cell migration into tissue.'

For Kutzler, this provides an explanation for a finding seen in research studies – that lymphoma, along with certain other cancers, occurs in much higher rates in neutered dogs.

'We know that there are a number of serious neoplastic diseases such as lymphoma, haemangiosarcoma, osteosarcoma that can develop after gonadectomy,' says Stefano Romagnoli, professor of small animal reproduction at the University of Padova; president of the Italian Society for Animal Reproduction; and chair of the World Small Animal Veterinary Association reproduction control committee.

'And we also know that some castrated dogs develop prostatic carcinoma. It's strange; counterintuitive. You would think that if you remove the testicles then

> **I understand this could be really frustrating for vets who were pretty much told: 'Neuter everything'**

the prostate would become atrophic. But for some reason, there is an increased risk of developing this prostatic neoplasia.'

Romagnoli does stress that the incidence of prostatic carcinoma in the general canine population is low. 'We're talking about perhaps a 0.5 per cent risk that in surgically castrated dogs may rise to 2 per cent. So we're really talking about small numbers, but it's the type of information that we should be conveying to owners.'

It is far from the only factor that leaves some experts concerned about the potential health impact of neutering. Research has also shown that urinary incontinence, obesity and hip dysplasia are all more common in castrated dogs. So too are immune conditions like hypothyroidism. 'We know that not all cases of hypothyroidism happen because dogs are neutered,' says Kutzler. 'But we also know, depending on what paper you read, that it's up to eight times more likely in neutered dogs than it is in intact dogs.'

These are health risks that hold true no matter the age at which a dog is surgically castrated, reason perhaps to reconsider the procedure as standard practice. But for one specific concern – bone health – there is at the very least an argument to delay the procedure.

'The gonadal hormones are needed to stop long bone growth length once it reaches the normally prescribed size,' explains Kutzler. 'So if you remove the gonads prepubertally, that delays when the growth plates will close and so those dogs that have their gonads removed prepubertally will be taller, they will have longer long bones.

'Among other things, that increases the risk for traumatic injuries, especially in their stifles and their elbows and their hips. That's because the angle of those joints is genetically determined based on the length the bone should have been, not the length it ended up being because the gonads were taken away too early.'

For Kutzler, the body of research means that surgical castration absolutely cannot be viewed as a zero-sum game. 'There is this idea that the procedure is done and the only health risks associated with it would be anaesthesia or some sort of surgical error. It's essentially said the procedure is safe. Okay, so the procedure itself may be safe, but the outcome of the procedure is not a zero-sum game. There are complications of removing the gonads and they have been well documented.'

It is incumbent upon the profession to consider alternatives to routine surgical castration

It also means, she says, that it is incumbent upon the profession to consider alternatives to routine surgical castration. While Kutzler strongly disagrees that neutering is the only solution to population control – 'look at the Scandinavian countries where neutering is either illegal or frowned upon; here we have countries where most of the population are not neutered and there is no dog overpopulation problem' – she acknowledges that for some owners controlling their animal's ability to reproduce may be appropriate.

'But my recommendation is to wait until after puberty before any sort of gonad removal surgery at all and to consider gonad-sparing options.'

In male dogs, those options might include vasectomy, using an implant that hormonally regulates reproduction, or intratesticular injection. The last of those possibilities gets a 'bad rap', Kutzler fears, but she believes it's unjustified.

'I really think that we need to reconsider the use of intratesticular injections in puppies. There have been two products that have really been evaluated quite thoroughly – zinc gluconate and calcium chloride. I think they get a bad rap because people use them improperly and in large, older dogs. They then have to use a large volume and have seepage and have to do a scrotal ablation.

'But if you're talking about puppies that have little tiny testicles, little peas, it's very easy to do an intratesticular injection in them. They're then sterilised but they're still going to make testosterone, so they're going to go through puberty fine.'

Ultimately, she believes that to consider such options is to uphold the fundamental principles of the veterinary profession – and that to surgically castrate unquestioningly is contrary to those principles.

'As veterinarians, we swore an oath to keep animals healthy. To create an unhealthy state intentionally or even unintentionally because of ignorance is, I think, against that oath.

'Where my passion lies now is trying to undo a message that has been ingrained in veterinarians – whether first-year veterinary students or [those who] have been practising for 40 years – that neutering your dog, without any reasoning or explanation of the risks, is the hallmark of responsible pet ownership,' she says.

Behavioural issues: castration may not be the solution

Removing the source of reproductive hormones may not be a panacea for unwelcome or unwanted behaviours – in fact, it may compound them.

WHEN Emma Hatton considers the behavioural issues surrounding surgical castration, she can't help but think back to a dog she rescued some two decades ago.

The animal had 'some difficulties around socialisation and social contact' but, in line with standard practice at the time, he was castrated once he reached about a year old.

'That was the expectation and it wasn't particularly questioned. And so that's what I did; I operated on him myself, in my own practice,' remembers Hatton, a veterinary behaviourist who is also a general practice vet. However, in hindsight, she thinks it wasn't necessarily the correct decision.

'He would have been a classic example of a dog that probably shouldn't have been castrated as early in his life as he was. Nowadays there would have been much more of a question mark over whether to castrate and, if a decision was taken to surgically castrate, when it should be done in terms of his physical and emotional maturity.'

Hatton sees this as an important shift in understanding in the veterinary profession. 'We are not only looking at older neutering, we are now very much questioning whether there should be any routine nature to it at all.'

Given that one of the major arguments traditionally presented for castration was a behavioural one, this is a change in approach that may seem confusing.

Owners have often believed that removing their dog's reproductive organs is likely to remove a range of behaviour they might consider undesirable. Some of this may relate to behaviour driven by reproductive hormones – the likes of urine marking, humping, mounting, roaming – but there is also a sense it promotes a much broader and vaguer behavioural change.

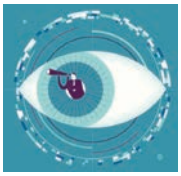
'I think there's still a lot of anecdotal talk of: "Castration will calm them down,"' says Hatton. 'All these sorts of phrases really do get bandied about a lot and it's not really anybody's fault – it's just what a lot of owners and vets will have grown up hearing.'

> **There's still a lot of anecdotal talk of: 'Castration will calm them down'**

The reality is that the impact of hormones on an animal's behaviour is simply not that cut and dried, she says. 'There is a small percentage of instances where perhaps we can be quite specific about how hormones are contributing to a certain behaviour. But so much more often it's a piece in a jigsaw – hormones are a contributing factor but need to be taken into account with a range of differential diagnoses for a particular behavioural presentation.'

This becomes particularly relevant when considering animals which display nervousness, fearfulness or aggression. In a dog that is reaching adolescence, such behaviour might traditionally have been interpreted as hormonal.

EXPERT VIEWPOINT



'Castration should not be considered standard procedure and non-neutering an exception. Neither of these two attitudes is, I think, correct. I think the correct attitude is to consider neutering as an individual choice that is applied to the behavioural and physical characteristics of an animal.'

Lorella Notari, *head of clinical behaviour, RSPCA*

'But actually the underlying emotion could be more one of uneasiness, fear or anxiety,' says Hatton. 'They may not want the other animal or a human to approach – they may feel threatened or a need to protect something of importance to them. They may communicate this by displaying a range of behaviours, some of which can progress to a level of aggression.'

'Aggression is not a label you can put on a dog,' argues Lorella Notari, head of clinical behaviour at the RSPCA. 'It's a sign of an internal state that can motivate certain behaviours. Sometimes owners interpret behaviours that lead them to have a certain kind of attitude towards a dog – if a dog growls in the presence of food or when there are toys, they think let's castrate. But those behaviours aren't abnormal.'

Furthermore, surgical castration is unlikely to resolve such behavioural presentations. There is some evidence, in fact, that removing testosterone prepubertally – preventing a dog from moving through a key development stage – might actually compound the problem.

'Sometimes we see an escalation of reactive behaviour towards people or other dogs during the adolescent period,' says Hatton.

> **It's been reported that anxiety, noise fear, fear of strangers does not improve with castration and may get worse**

'But it's a development period and their confidence levels will fluctuate a lot during all their different development phases, and in adolescence their hormone levels [will fluctuate] too.

'So if we're going to go in and dramatically change hormones at that time, we've got to be very aware of the impact that can have. An animal might actually benefit from the contribution of hormones through that period and into their maturity.'

Notari recalls a case in which an unconfident, nervous, fearful animal was surgically castrated despite the vet having warned the procedure would not address those behaviours. 'The owner thought, okay, but it will probably be calmer. But that's not what happened. It became more anxious. It has been reported by a few authors now that anxiety, noise fear, fear of strangers does not improve with castration and may get worse.'

There are even questions over whether apparently sexually driven behaviours can always be reduced by surgical castration. 'Behaviours in male dogs such as urinary marking and humping or mounting can be quite classically thought of as hormonally driven,' says Hatton. 'But we can also see some of those behaviours in female dogs, and sometimes with males they can be performed in a particular set of circumstances.

'It might be a displacement behaviour – like us twiddling with our hair, having worry beads in our pocket, or chewing our nails. So we have to ask if these behaviours are entirely hormonally driven or if it's just something the dog does as one of its repertoire of behaviours, and whether it might just be a coping strategy in a particular situation.

'From a behavioural point of view, there are still so many questions,' Hatton concludes. 'And it's shades of grey – certainly not black and white.'

Some 20 years on from surgically castrating that nervous rescue dog at 12 months old, believing it was best practice and in his best interests, she has a different perspective on the procedure.

'Now I don't have a set timescale for castration by age or breed or anything like that. I just try to treat each dog as an individual, and work out from the clients what information they have had and where they are coming from. Are they looking at castration more from the health side, or more on the behavioural side? And have they been suitably informed to make a decision?' ●

'A LOT OF OUR CLIENTS HAVE BEEN CONCERNED ABOUT BEHAVIOURAL PROBLEMS' – THE VIEW FROM DENMARK

Kathrine Thejll Kirchhoff is a European veterinary Specialist in animal reproduction working at a referral clinic in Denmark. She is one of the authors of the Danish Veterinary Association's guidelines on neutering, which recommend that no dog should be castrated before puberty.

'A lot of clients are now going to the internet, searching for articles and asking us a lot of brilliant questions about whether it is right to castrate a dog at all.

'Many have been aware that castration can be an issue with behavioural problems. If you castrate dogs that have been nervous, maybe aggressive, and do it too early we know it can definitely make the issue worse. That's something a lot of our clients are concerned about.

'A group of specialists wrote the Danish Veterinary Association guidance. We had an ecologist, someone working with behaviour problems, and we looked into all the literature that has been published.

'The conclusion we came to was that male dogs should never be castrated before puberty. To make things easier for owners, we've recommended that means not before one year of age.

'Sometimes whether to castrate depends on what kind of family and what kind of environment the dog is living in. Owners might worry that if there are intact bitches living nearby, their male dog might run around.

'But in the last couple of years, I can't remember having had a client who was very eager to castrate before puberty.'

Charities consider their castration policies

While population control remains a key priority for the UK's leading dog welfare charities, it must increasingly be balanced with concerns about the possible health impact of castration.

AROUND a year ago, the RSPCA made a small, subtle and yet significant change to its policy on neutering.

'Before, our policy was essentially that every dog [under our care] must be surgically castrated, without question,' says Caroline Allen, the society's chief veterinary officer. 'Now we've introduced nuance into that, really recognising that castration can cause harm.'

Lorella Notari, the charity's head of clinical behaviour, explains: 'Where there are dogs with behavioural issues, especially related to signs of generalised anxiety, we normally now suggest waiting to castrate to help the dog settle when they are rehomed.'

The result is that, in certain situations, the organisation is now rehoming dogs without them having been surgically castrated. For Allen, it's an evolution that sensibly reflects changing knowledge on the potential impact of castration. It is, she says, 'right' to have a discussion about the issue.

Even so, it's a conversation – and a related policy change – that she believes not so very long ago would have been seen as unthinkable: the UK's largest animal welfare charity changing its position on a procedure traditionally seen as key to population control. Yet Allen anticipates the organisation will soon go further still.

'I think we're likely to push back the dates of neutering for dogs expected to be over 25 kilos,' she says. 'I think for larger breed dogs, there is evidence that there are health considerations when castrating. There is more evidence about degenerative joint diseases [being potentially linked to early castration] and those are significant conditions. They can have a significant impact on quality of life.

'I think the health benefits of neutering in male dogs have never been massively strong,' she adds. 'With things like benign prostatic hyperplasia or adenocarcinomas, generally if they're caught early then castration at that time will have the desired effect. I think avoidance of prostatic disease is sometimes used as a justification for castration but, actually, for me the causes of all the different prostatic diseases that a dog can get are complex.'

While these sorts of deliberations may be relatively new within organisations focused on population control as key to welfare, the RSPCA is not unique in having them. In July this year, Dogs Trust made a similar shift to its policy.

Rachel Groarke, one of the charity's veterinary surgeon managers, emphasises that population control is 'very important' to Dogs Trust. 'But it's also important for us to be evidence-based in everything we do.'

That's why, following a literature review, the organisation's position on neutering has changed. Previously it neutered all dogs under its care at four to six months of age, unless there were medical or behavioural reasons not to do so. For any dog with a predicted adult body weight of up to 20 kilograms, this approach still stands.

'However for dogs with an expected weight of 20 to 40 kilos, we're now waiting until they're 12 months of age before we castrate them. And for any dogs that are expected to be over 40 kilos, we're waiting until 18 months of age.'

Groarke is clear that the need for population control has not ceased. In fact, she says, Dogs Trust has recently seen a growth in the relinquishment of pregnant dogs and of litters of puppies. 'From the

> **For larger breed dogs, there is evidence that there are health considerations when castrating**

start of the year to the end of August there's been a 30 per cent increase in the number of puppies born on site in comparison to the same time period last year. That implies that we have had more pregnant mums handed over this year.

'There has also been an increase in the number of dogs handed to us because they were part of an unwanted or unplanned litter, and the proportion of handovers with that stated as the reason has increased too.'

The change in neutering policy is therefore an attempt to balance the need to limit the number of unwanted animals with 'the potential negative impact of castration, particularly on skeletal maturity and joint health'.

There's another consideration for animal welfare organisations too: that surgical castration as standard practice across the UK veterinary profession may in some instances have served to control the wrong parts of the animal population.

Before joining the RSPCA, Allen spent 20 years as a practising vet. That included time at a practice which 'really pushed neutering; we absolutely believed it was the right thing to do'. It meant that when owners raised the possibility of breeding from their dog, her inclination was to raise reservations.

'I think because of the sheer overpopulation in the past, we as a veterinary profession shut down those conversations – "There's too many dogs, please don't do it. Neuter. That's the right thing to do."'

Now, however, she is concerned that there was an unintended consequence of that sort of advice – namely what she describes as 'massive problems' with dog breeding as it stands at present.

'But it isn't as straightforward as a numbers game. It's around who is choosing to breed, how they're choosing to breed, the breeds that they are breeding, the motivations for breeding.

'I think actually we do need some of the sort of breeding we had in the past, of a family having a dog and deciding to breed from it in a responsible way. Because at the moment there is a shortage of good, suitable, well-bred, well-socialised puppies – which is why there is a growth in importing animals, with all the challenges that can bring. If we look at the population of puppies out there against what we would like people to be owning, it's quite a mismatch.'

Which dogs would we really like not to be neutered, so that they might be used for responsible breeding?

She says the question becomes 'which dogs would we really like not to be neutered, so that they might be used for responsible breeding'.

That will mean not only that neutering becomes a decision that has to be made on a pet-by-pet basis ('and that is harder for practices – and certainly even more difficult for a charity, where people want us to come out with very clear policies') but also that the profession may need to come to terms with potential previous harm.

'It is quite uncomfortable to confront that as a profession we may have caused some harms with the advice to neuter as standard. I think all we can do is say, well, we really gave that advice in good faith. We didn't do it for the wrong reasons. We weren't doing it because we wanted to do more procedures. We really did it with best interests at heart.

'But now we have this new evidence, I think we do have to stop, think and readjust based on that new evidence, even if that is a bit uncomfortable. Certainly decisions over neutering are more complex than they used to be,' Allen concludes.

EXPERT VIEWPOINT

'Responsible breeding is not avoiding random dogs mating with each other and not having dogs on the streets and not having unwanted puppies. It's actually making sure that people are breeding from the right dogs with the right temperaments and the right traits that you are wanting to bring into this world. There are some dogs that we might not want to be breeding from because they aren't particularly good examples of the breed from a behavioural point of view. That doesn't mean those dogs should automatically be neutered to stop them breeding though, because there is a lot more human control over breeding now than there might have been a couple of generations ago.'

Emma Hatton, *veterinary behaviourist*

An owner's perspective

Bailey the German shepherd dog came into Ellen's life in early 2022 when he was two years old. Her initial intention was to have him surgically castrated. But her own research, coupled with support and advice from her vets, ultimately made her decide against it.

'BAILEY came to us as a rehome as his previous owners' situation had changed and they could no longer keep him.

'He is my first dog, although my partner has had dogs before and I have experience with them. We were waiting until the time was right before we got one of our own.

'I was planning on getting Bailey castrated as one of the first things I did. It was seen as the responsible thing to do, after all. However I am a scientist and so did my research first.

'I saw that for large breeds it was advised to wait until they are fully grown before surgically castrating. I also discovered that surgical castration is not always seen as the desired thing to do now, depending on your lifestyle and the individual dog. So I decided that until he had settled in I was going to leave him intact.

'Bailey hadn't been walked so we discovered the outside world was the biggest distraction for him. That's no disrespect to his previous owners – they had acres of land for him to run around in, but it meant he didn't know leash manners at all. He had little to no engagement with me and his nose was plastered to the floor almost constantly.

'And it was soon apparent that, being a Covid puppy, Bailey had missed out on important socialisation. Although thankfully wonderful with people and children, his attitude towards other dogs was less than desirable. We'd not been made aware of that when we rehomed him, so it came as a large shock.

'In regards to neutering him, this gave us a dilemma. What sort of reactive was he? Aggressive? Fearful? Frustrated? And how would his loss of testosterone affect that? I certainly didn't want to do anything irreversible. My concern was that surgically neutering him could cause him to lose his confidence due to the decrease in testosterone. This could be disastrous for a reactive dog.

My vets gave me unbiased advice about our options

'My vets were invaluable here. They didn't force me into any choices and gave me unbiased advice about our options. Ultimately we decided to go for medical castration. I am not going to say it's the right choice for every dog but we feel it has been for ours. His engagement is much better now that "lady smells" are less appealing to him, and that's enabled us to work on our training with much better results.

'It seems his previous reactivity was thankfully not fear based or aggressive and purely due to frustration. We were recently able to go to a charity dog show and look at the stalls with minimal outbursts which is something that would have been impossible a year ago.

'Upon speaking to other dog owners, I have discovered that most people haven't heard of alternatives to surgical castration. I have even had several owners upset because they would have made the same choice I had if given the option, especially in cases where surgical castration has altered their dog's behaviour in a negative way. Many people rely on their vets for this kind of advice and won't be doing their own research; there is a lot of trust placed in vets in this regard.

'Overall I am happy with my decision for my dog, but I am not against surgically castrating him in the future if needed.'

Vet Record

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BEHAVIOURAL MEDICINE AND ANIMAL WELFARE
27th-29th September 2018, Berlin

Influence of gonadectomy on canine behaviour

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In veterinary practice, surgical sterilization of dogs is one of the most common surgical procedures performed and it has been considered for decades to be a routine standard for the prevention of numerous undesirable behaviours, medical conditions, and diseases [1,2]. Because of the aforementioned variables, there currently is no single recommendation regarding gonadectomy that would be appropriate for all dogs [3] .
Aim of the current study was to explore relationships between gonadectomy and various canine behaviors. Data were collected over time before and after surgery from owners of dogs sterilized and from owners of healthy entire dogs to compare behaviors between the two groups.

The study was carried out on 156 clinically healthy dogs (78 control group and 78 experimental group balanced for gender). Dogs' owners were asked to complete an initial questionnaire (time 0). A follow-up questionnaire, including dog's demographic information and behavioural history, as well as information on the dog's physical and social environment, was sent at 10 days, 1, 3, 6 and 9 months after the first examination. Answers to the questionnaire were scored, descriptive statistics were calculated to provide a general description of the two experimental groups. Any differences in behavior, management and environment between the two groups, were compared using a Pearson Chisquared test. Differences were considered to be statistically significant if $p \leq 0.05$. A Wilcokson test was used to investigate potential differences in dogs' behavior between time periods.

Fig 1. SLEEPING BEHAVIOUR IN THE EXPERIMENTAL GROUP DURING TIME

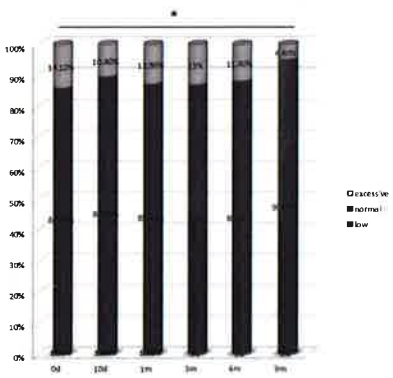


Fig 2. OWNER AGGRESSION IN THE EXPERIMENTAL GROUP DURING TIME

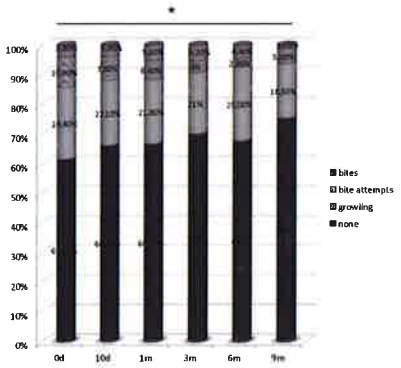


Fig 3. ROAMING BEHAVIOUR IN THE EXPERIMENTAL GROUP DURING TIME

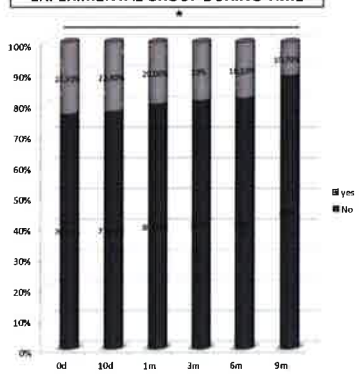
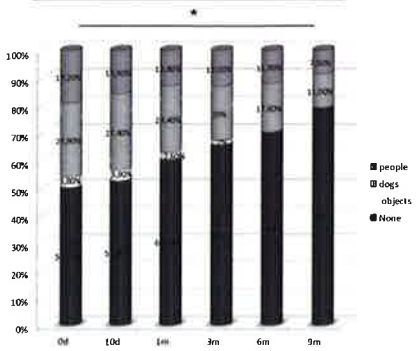


Fig 4. MOUNTING BEHAVIOUR IN THE EXPERIMENTAL GROUP DURING TIME



Answers of the questionnaire revealed that no change was found between time 0 and 9 months in both groups of dogs for eating and drinking behaviours, exploration, grooming, chewing objects, destruction, excessive barking, pica and coprophagy. No difference was found also for inappropriate elimination in the house and repetitive behaviours. In the experimental group sleeping behaviour increased significantly ($p \leq 0,05$) over time (fig.1), while no change was observed in the control one. Owner aggression (fig.2), pull on a leash, roaming (fig.3) and mounting behaviours (fig.4) decreased significantly in the experimental dogs between time 0 and 9 months ($p \leq 0,05$), while no change was observed in the control group. Only a tendency was detected for inter-dog aggression.

The necessity, and in particular the timing, of gonadectomy in dog is still controversial. This is mainly because gonadectomy confers a mixture of benefits and adverse effects that depend upon the age at neutering, sex and breed. Our study diverges with some previous findings. More research is needed to contribute to the debate regarding whether and how gonadectomy can affect behaviour.

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Eindhoven, The Netherlands
26th - 28th September 2019

Spotlight on animal behaviour problems

Focus on welfare, ethical and legal aspects

Proceedings of the 2nd Annual Meeting of the European Veterinary Congress of Behavioural Medicine and Animal Welfare (EVCBMAW)

IS GONADECTOMY IN DOGS AND CATS REALLY IMPROVING WELFARE?

Titeux E.[1], Gilbert C.[1,2]

Gonadectomy in dogs and cats is widely used to prevent unwanted litters, as well as cancers. Gonadectomy is also performed for the resolution of behaviour problems particularly aggressions (Overall 2001). Gonadectomy is often seen like a panacea with few side effects counterbalanced by multiple obvious benefits (Spain 2004). However recent publications temper this enthusiastic point of view (Farhoody 2018). Neutered animals could deal with several side effects which might impair their welfare. The aim of this presentation is to offer a new insight into companion animal gonadectomy.

In female cats, gonadectomy is nowadays the best means to prevent unwanted litters and mammary carcinomas. For the male, castration decreases roaming and mating behaviours, thereby limiting wounds, diseases and roads traffic injuries. However in feral cats, a vasectomy and hysterectomy programme could lead to a faster decrease in the cat population compare to a tra-neuter-return programme (McCarthy 2013). Therefore in cats, the issue should be more focused on the timing of gonadectomy, i.e. pediatric versus juvenile (Porter 2014). In dogs, gonadectomy is used to prevent pyometra, mammary tumors and roaming. Nevertheless negative effects include weight gain, loss of social behaviours with unpredictible effects on aggression (Farhoody 2018). Moreover neutered dogs are more affected by several cancers. (Hart 2016). Their cognitive function is also impaired both early and late in life (Hart 2001, Scandurra 2019).

In consideration of animal welfare, we should be more cautious concerning the gonadectomy of companion animals. A case by case recommendation seems to be more suitable.

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animals

MDPI

Review

Desexing Dogs: A Review of the Current Literature

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Simple Summary: Desexing is a general term for interventions suppressing fertility in dogs, most commonly by surgically removing the testes or ovaries ("gonadectomy"). Desexing is promoted for population control, health benefits, and behavior modification. Surprisingly, the existing evidence shows no effect of desexing on population size in companion or shelter dogs; however, an effect has been shown for desexing female free-roaming dogs. Desexing has consistently been shown to change various health risks, including a reduction in pyometra and mammary tumor risk, as well as an increased risk of cranial cruciate ligament rupture, several forms of cancer, and obesity in both sexes. Other health effects vary considerably between breeds and sexes. A lifespan advantage in desexed dogs has consistently been shown in females, while the evidence is inconsistent in males, and the effect is smaller in studies that found one. There is more literature on behavioral effects in males than in females, and the evidence suggests reduced libido, roaming, conspecific mounting, and urinary marking in a large percentage of gonadectomized males, and reduced male dog aggression in a majority of males gonadectomized because of behavioral problems. The decision whether to desex dogs needs to be individualized based on the available evidence.

Abstract: *Background*: Desexing dogs is promoted for population control, preventative healthcare, and behavior modification. Common methods are orchiectomy and ovariectomy/ovariohysterectomy. GnRH superagonist implants are available in some areas. Alternative methods like vasectomy and salpingectomy/hysterectomy are uncommon. The terminology used to describe desexing is inconsistent and contradictory, showing a need for the adaption of standardized terminology. *Population Control*: Surprisingly, empirical studies show no effects of desexing on population control in companion and shelter dogs despite desexing being consistently recommended in the literature. There is evidence for a population control effect in free-roaming dogs, where desexing also has benefits on zoonotic disease and bite risk. Population control in free-roaming dogs is mostly correlated with female, not male desexing. *Health and Lifespan*: Desexing affects numerous disease risks, but studies commonly neglect age at diagnosis and overall lifespan, age being by far the most important risk factor for most diseases. We argue that lifespan is a more important outcome than ultimate cause of death. A beneficial effect of desexing on lifespan is consistently demonstrated in females, while evidence for a beneficial effect in males is inconsistent. Studies are likely biased in desexing being a proxy for better care and desexed dogs having already lived to the age of desexing. Desexing reduces or eliminates common life-limiting diseases of the female reproductive system such as pyometra and mammary tumors, while no analogous effect exists in males. Disease risks increases across sexes and breeds include cruciate ligament rupture, various cancers, and obesity. Urinary incontinence risk is increased in females only. Various other disease risk changes show considerable variability between breeds and sexes. *Behavioral Effects*: Desexed males show reduced libido, roaming, conspecific mounting, and urinary marking frequency, as well as reduced male dog-directed aggression in a majority of males desexed for behavioral reasons. There is a detrimental effect on the risk and progression of age-related cognitive dysfunction. Desexed dogs may be less likely to cause bite injuries across sexes. The evidence for other effects such as human-directed aggression, human or object mounting,

resource guarding, or shyness and anxiety is inconsistent and contradictory. There are few studies specific to females or individual breeds. *Conclusions*: The evidence for a beneficial effect of desexing is stronger in female than in male dogs; however, there is significant variation between breeds and sexes, and more research is needed to further elucidate these differences and to arrive at individualized evidence-based recommendations for clinical practice.

Keywords: gonadectomy; desexing; orchiectomy; ovariectomy; ovariohysterectomy; spaying; neutering; dogs; epidemiology; lifespan; disease risk; population control; behavior; review

1. Introduction

Surgical desexing, usually by means of gonadectomy, is the most widely performed surgery in dogs and is commonly promoted as being part of responsible dog ownership in many English-speaking countries as well as parts of non-anglophone Western Europe. Desexing prevalence reaches 64% across both sexes in the United States [1], with some local jurisdictions having adopted mandatory desexing laws [2]. Desexing is somewhat less common in anglophone Western Europe, with prevalences reaching 54% in the UK [3] and 47% in Ireland [4]. In contrast, desexing is considered illegal without a clear medical indication in Germany [5] as well as some of the Nordic countries [6–8], with prevalences below 10% in Sweden [9].

A large variety of methods for desexing dogs has been described in the literature. While we attempt to give a complete overview of these methods, it is worth pointing out that gonadectomy is by far the most common form of desexing currently used in practice across the world, and the literature on its various consequences reflects this fact in that it predominantly focuses on the consequences of gonadectomy as opposed to other methods of desexing. Therefore, most of the evidence discussed in this review will necessarily be specific to gonadectomy as opposed to desexing in general.

Justifications for desexing put forward by the veterinary and shelter communities can generally be divided into three main lines of argument. These are (1) population control, (2) health benefits, and (3) behavioral benefits. For the purpose of this review, we will address all three of these aspects in separate sections.

2. Terminology

A wide range of terms has been used to describe the various methods of desexing, and no set of standardized terminology exists. For instance, both "neutering" and "castration" have been used by some authors as a sex-specific term for male gonadectomy, while others have used them as sex-neutral terms for gonadectomy in both sexes. "Sterilization" has been used as a sex-neutral term for gonadectomy, a sex-neutral term for methods that induce infertility without using gonadectomy, and a sex-specific term for female gonadectomy. In contrast, "spaying" is consistently used as a sex-specific term for female gonadectomy; however, it may or may not also indicate concurrent hysterectomy and has occasionally also been used to denote hysterectomy without ovariectomy or with an ovarian autograft ("ovary-sparing spay").

In the interest of consistency and clarity, for the scope of this review we will use "desexing" as a sex-neutral term for any procedure intended to cause temporary or permanent loss of fertility, "gonadectomy" as a sex-neutral term for the surgical removal of the gonads in both sexes, "orchiectomy" as a sex-specific term for gonadectomy in males, "ovariectomy" as a sex-specific term for gonadectomy in females, and "ovariohysterectomy" as a sex-specific term for gonadectomy combined with hysterectomy in females. Specific methods not involving orchiectomy or ovariectomy will be referred to using their commonly used names as taken from the literature.

3. Desexing Methods

In the broadest sense, desexing methods can be divided into surgical and non-surgical procedures [10–12]. Given the significant anatomical and physiological differences between male and female dogs, we will further distinguish between methods used in males and methods used in females as appropriate.

3.1. Surgical Methods

Surgical methods are the most widely used form of desexing in dogs, and are the most commonly performed surgeries in the species overall [7,10,13–16]. They can be differentiated into surgeries that remove the gonads (orchiectomy, ovariectomy, ovariohysterectomy) and surgeries that do not (vasectomy, salpingectomy, hysterectomy), the former methods being considerably more common than the latter.

3.1.1. Orchiectomy

Orchiectomy (also called orchidectomy) is the surgical removal of the testes in male dogs. Two main surgical methods exist to achieve this goal, which are referred to as the open and the closed technique. In both techniques, the patient is placed in dorsal recumbency and the surgeon places a skin incision cranially from the scrotum through which the testicles are extracted. In the open technique, the surgeon then also opens the vaginal tunic before placing a ligature around the vas deferens and testicular artery and removing the testicle; in the closed technique, the vaginal tunic is left in place and a ligature is placed around the tunic as well as the testicular artery, cremaster muscle, and vas deferens inside the tunic before removing the testicle and surrounding structures distal of the ligature. In addition to these two primary methods, a sutureless scrotal approach has recently been described for use in juvenile male dogs [17], and testicular implants for orchiectomized dogs are also commercially marketed [18].

Short-term complications of orchiectomy are rare and are mostly limited to the usual non-specific surgical risks, i.e., anesthetic complications, swelling, hematoma, bleeding, and infection [10].

3.1.2. Alternative Surgical Methods in Males

Various surgical desexing methods not involving orchiectomy have been proposed for male dogs; however, none of them have been widely adopted in practice. The most common such surgery is vasectomy, which consists of the surgical transection of the vas deferens, leading to azoospermia and thus, infertility without affecting testosterone levels [19]. Methods include open [20,21] and endoscopic vasectomy [22], as well as experimental methods including implantation of an intravasal filtering device [23], and non-scalpel methods involving ultrasound [24], laser surgery [25,26], and electrocoagulation of the vas deferens [27].

Vasectomy does not result in changes in circulating testosterone levels, libido, or other male-specific behaviors, and as such has been suggested for use in free-roaming and/or feral dogs as a potentially more effective means of population control than orchiectomy [28–30]. Specific complications of vasectomy appear to be rare in dogs and can include spermatocele [31], sperm granuloma [31,32], and testicular degeneration [33].

In addition to the various forms of vasectomy listed above, surgical methods described in the literature also include the Burdizzo castrator [34], while use of the elastrator is considered inhumane in dogs [35].

3.1.3. Ovariectomy and Ovariohysterectomy

Ovariectomy (also called oophorectomy) is the surgical removal of the ovaries in female dogs, while ovariohysterectomy refers to the surgical removal of both the ovaries and the uterus [10]. Ovariohysterectomy is generally considered the method of choice in older females. In North America, it is also the most commonly used desexing procedure in young females, while ovariectomy without

concurrent hysterectomy is generally considered the method of choice for desexing young females in Western Europe [36,37].

In contrast with orchiectomy, both ovariectomy and ovariohysterectomy require a laparotomy in order to access the surgical site. There are two commonly used surgical approaches for both procedures in female dogs, which are the midline incision and the flank incision. The midline incision is the more commonly used approach of the two, while the flank incision is mostly performed in shelter medicine settings, in small or narrow-bodied dogs, or in dogs with significant mammary development [37,38].

The midline incision is performed with the patient in dorsal recumbency and is placed in the linea alba slightly caudal of the umbilicus. The flank incision is performed with the patient in lateral recumbency, with the right flank approach being more commonly used due to anatomical considerations [38]. The incision site lies caudal of the midpoint between the iliac crest and the last rib, with the abdominal muscle being dissected bluntly. Both methods then use isolation and ligation of the ovarian pedicle and removal of the ovaries, followed by ligature of the uterine body and cervix and removal of the uterus in case of ovariohysterectomy.

Laparoscopic ovariectomy and/or ovariohysterectomy techniques (sometimes including electrocoagulation) have also been described for female dogs, and may be associated with fewer signs of post-operative pain and faster recovery times; however, they are also associated with higher cost [39–41].

As is the case in orchiectomy, short-term complications of ovariectomy and ovariohysterectomy are rare and mostly limited to the usual non-specific surgical and anesthetic risks as listed above [10].

3.1.4. Alternative Surgical Methods in Females

Alternative surgical desexing methods in female dogs are not widely used in practice and include salpingectomy, i.e., the surgical transection of the fallopian tubes [42], hysterectomy without ovariectomy, and ovariectomy or ovariohysterectomy combined with an ovarian autograft [43,44]. Their medical consequences are not well studied. As is the case in vasectomy, these methods do not change the hormonal status of the dog, meaning that the health consequences can be expected to be similar to these found in intact individuals inasmuch as the susceptible organs remain in the body.

3.2. Non-Surgical Methods

Various non-surgical methods to induce infertility and/or suppress libido and other hormone-mediated aspects of behavior in dogs have been described in the literature. They can roughly be divided into hormonal methods, immunological methods, and chemical or physical sclerotization methods [11]:

3.2.1. Hormonal Methods

The most commonly used hormonal methods have used progestins to induce reversible infertility in both male and female dogs and to also suppress both libido and estrus, as well as other specific effects of testosterone and estrogens. They include competitive steroid hormone inhibitors such as cyproterone acetate [45], as well as progestogens such as megestrol acetate, medroxyprogesterone, and proligestone [46]. Suppression of estrus and fertility is readily achieved in female dogs using these methods; however, they have been shown to be less reliable in achieving infertility in male dogs [47]. Exogenous testosterone has also been used to successfully suppress fertility in both male and female dogs; however, its undesirable androgenic side effects render it impractical for clinical use [47].

More recently, GnRH superagonists such as deslorelin applied in the form of slow-release subcutaneous implants have been used to successfully induce reversible infertility lasting several months to years in both male and female dogs. After an initial gonadotropin surge that in females may induce fertile estrus, these implants suppress the release of gonadotropins in the pituitary gland, which results in downregulation of the gonads mimicking the effects of gonadectomy, and resulting in complete infertility and undetectable levels of sex steroids within 6–8 weeks after implantation [48–50].

While these implants are not currently commercially available in North America, they are now widely used in veterinary practice in both Europe and Australia. Their use has also been suggested to delay the onset of puberty in both male and female dogs by several years [51–53].

Phytoestrogens have also been suggested as a possible method of inducing infertility in male dogs; however, this has not been applied in clinical practice thus far [54].

3.2.2. Immunological Methods

There have been several immunological methods investigated to achieve infertility in both male and female dogs; however, while there have been some promising results in studies on free-roaming dogs [55,56], none of the studied products have thus far been adopted in clinical practice. Proposed and tested targets for immunization have included sperm antigens, the zona pellucida of the oocyte, GnRH, and LH [56–62].

3.2.3. Sclerotization Methods

Sclerotization methods have thus far only been used on male dogs, have used both chemical and physical means of achieving sclerotization, and have been aimed at either the testicle, the epididymis, or the vas deferens.

Various chemical sclerotizing agents have been proposed, among them intratesticular injections of calcium chloride or zinc gluconate [63–65], intraepididymal injections of chlorhexidine gluconate, zinc arginine, or formalin [28,66,67], and injections of ethanol, silver nitrate, acetic acid, formalin, sodium tetradecyl sulfate, sodium morrhuate, or potassium permanganate into the vas deferens [68]. Among these methods, intratesticular injections of calcium chloride or zinc gluconate have seen limited use in private veterinary practice and particularly in shelter medicine.

Physical sclerotization has used ultrasound application aimed at either the whole testicle [69,70] or the epididymis [24], but has not been used in clinical practice thus far.

4. Population Control

Starting in North America during the 1970s, concerns about pet overpopulation and implementing population control measures have historically been at the forefront of desexing advocacy originating from both the veterinary profession and the shelter community, focusing on both animal welfare and public health aspects associated with free-roaming dogs [71–75]. However, considering both the substantial body of literature advocating in favor of desexing as a means of population control and the widespread use of desexing in shelters and by private practice veterinarians over the last 40–50 years [3,13,14,76–79], the body of evidence investigating the effectiveness of desexing to actually achieve population control in companion and shelter dogs is surprisingly slim [80,81], and the evidence from such studies does not generally support the existence of an effect of desexing programs on population control, as discussed below.

Another concern in both desexing advocacy and the study of its effects on population control is a lack of sex-specific considerations. In mammalian population dynamics, it is generally accepted that reproductive rates in a population predominantly depend on the availability of fertile females, while the number of fertile males does not directly affect population size at the desexing rates that are achievable in practice, indicating that inhibiting female reproduction is the intervention that will predominantly determine outcomes of desexing interventions [55,82–84]. However, these considerations have not been incorporated into the currently recommended protocols [13,77,79] and are largely absent from the veterinary literature.

In summary, the available empirical evidence does not support the notion that gonadectomy programs have an effect on population size in companion and shelter dog populations, while there is evidence for such an effect in free-roaming dog populations. Demographic surveys suggest that this lack of an effect in companion and shelter dog populations may in part be due to the fact that at least in the US environment, roughly two thirds of dog litters are intentionally bred rather than accidental [85].

There is also evidence that dogs being relinquished to shelters may be more strongly correlated with the existence of behavior problems than with population size. In addition, the lack of a specific focus on females that is common in companion dog gonadectomy programs may also contribute to the lack of an effect on population control.

The existing literature on the effects of desexing on population control in dogs can be divided into studies based on mathematical modelling, empirical studies on companion and shelter dog populations, and empirical studies on free-roaming dog populations.

4.1. Studies Using Mathematical Modelling

Mathematical modelling studies use various assumptions about a modelled dog population based on empirical surveys of existing dog populations, and then use these assumptions to test the influence of various variables and interventions on birth rates, mortality rates, as well as immigration from and emigration into adjacent populations in order to estimate the impact on population dynamics and ultimately population size over various lengths of time.

Such studies have thus far mostly been limited to scenarios using free-roaming dogs, or scenarios where a significant number of such dogs exist alongside the population of owned companion dogs. The most important factors identified using such studies are generally the percentage of fertile females [82,83], the carrying capacity of the environment [83,86,87], a sustained desexing effort over time as opposed to a one-time intervention [83,86], and the introduction of intact animals from adjacent populations and/or through animal abandonment [86].

4.2. Empirical Studies on Companion and Shelter Dogs

Empirical studies on the effects of gonadectomy on companion and shelter dog population dynamics exist from the United States, Canada, and Brazil. These studies generally investigate populations over various points in time before and after an intervention was implemented, or compare areas where an intervention was implemented to similar areas without such an intervention. Methods used include owner surveys as well as records taken from private and public animal shelters, and the variables analyzed generally include shelter intake and euthanasia rates over time. Only one study published thus far has considered existing population dynamics trends preceding the time at which the intervention was introduced [88].

All studies in companion and shelter dogs thus far have examined the impact of gonadectomy across both sexes and have not considered sex-specific effects. With one exception, the available empirical studies have not found an effect of gonadectomy programs on common measures of population size in companion or shelter dogs.

Analysis of data generated from the "Maddie's Fund" initiative promoting low-cost gonadectomy programs in the US States of Utah and Alabama, as well as in Maricopa County, AZ, Alachua County, FL, and the city of Lodi, CA found that these programs significantly increased the percentage of gonadectomized dogs in the population, but did not affect shelter intake rates [81].

A study in two populations in New Hampshire and Austin, TX studied shelter intake and euthanasia numbers before and after the implementation of a gonadectomy intervention; in New Hampshire, following the introduction of a "neutering initiative" promoting low-cost gonadectomy in both sexes, a beneficial effect was found in cats, but not in dogs. In Austin, a free gonadectomy program was introduced in specific areas of the city and was shown to slow the increase of dog intake and euthanasia rates in the targeted areas as compared to areas where no such intervention had taken place [89].

A study in Transylvania County, NC, examined the impact of a subsidized gonadectomy clinic on shelter intakes and euthanasia before and after the clinic opened. In this study, the shelter intake rate had already been decreasing before the clinic had opened, and the decrease did not accelerate afterwards. Euthanasia rates had also been decreasing before the clinic had opened; however, they stabilized afterwards [88].

A study in Santa Clara and San Mateo Counties in California found that the introduction of gonadectomy vouchers by the shelter system resulted in a reduction in animal shelter intake numbers in cats, but not in dogs [90].

In one study, a door-to-door outreach campaign promoting gonadectomy in an underserved area of New York City had no significant effect on the rates of animal abandonment compared to a control population [91].

In a study of two Native American communities in North Saskatchewan, Canada, consisting of an environment where both owned companion dogs and unowned free-roaming dogs were present, introducing a subsidized gonadectomy program for companion dogs combined with capture and gonadectomy of free-roaming dogs resulted in a significant increase in the percentage of gonadectomized dogs, a decrease in the number of litters observed, and a decreased population size nine months after the intervention; however, the decrease in population size was caused by placing unwanted dogs that were identified during the intervention with households outside the communities [92].

Finally, in one study in the Brazilian city of Curitiba, it was found that a three-year population management and control program promoting gonadectomy to companion dog owners did not significantly affect the number of dogs in the city [93].

4.3. Empirical Studies on Free-Roaming Dogs

Empirical studies on free-roaming dogs have largely been conducted in environments where dogs reproduce outside of human interference and where the presence of free-roaming dogs poses significant risks to human health, either as disease reservoirs for infections such as rabies [56,94,95], or by causing a significant number of bite injuries [96]. In these situations, free-roaming dogs are generally trapped, desexed using gonadectomy or non-surgical methods, vaccinated against rabies and other diseases, treated against infections and common parasites, and then released [56,94–98].

In one study in Jaipur, India conducted between 1994 and 2002, it was estimated that 65% of all female dogs were gonadectomized during this period, which resulted in a decrease in the free-roaming dog population size of 28% over the period [94]. Another study conducted before and after implementation of a gonadectomy/rabies vaccination program in Jodhpur, India found a significant decline in the dog population size in three out of five surveyed areas following implementation of the program. A different study conducted in several Indian cities in 2011 found that gonadectomy also resulted in improved body condition scores and reduced disease and parasite load (other than ticks) in gonadectomized dogs as compared to intact controls [97].

In this context, it has also been suggested that vasectomizing rather than gonadectomizing male dogs may exert a beneficial effect in that these infertile males will continue to compete with fertile males for the available fertile females, thus reducing the rates of successful impregnation in the latter [30]. To our knowledge, this hypothesis has not been tested in practice thus far.

5. Lifespan and Disease Prevalence

5.1. Lifespan

As we summarize below, much has been published on various disease risk changes associated with desexing or not desexing; however, most of these studies did not take lifespan or age at diagnosis into account. This is problematic for two reasons: First, age is by far the most important risk factor for a wide variety of diseases in dogs, including many of the diseases for which a correlation with desexing has been described [99–102]. A change in individual disease prevalence may therefore simply indicate a difference in lifespan. Second, while desexing or not desexing will indeed change a dog's risk of developing various life-limiting diseases, it ultimately does not prevent death. This may seem obvious and even trivial when stated in this manner; however, a large part of the literature appears to neglect this fact and focuses largely on causes of death. Given that lifespan is finite in both intact and desexed dogs, we argue that more emphasis should be placed on lifespan (and, ideally, healthspan) as

a relevant measure of positive or negative health outcomes associated with desexing dogs, regardless of ultimate cause of death.

While studies on individual disease risk changes associated with desexing are abundant, the number of studies examining the effects of desexing on lifespan is more limited [103–108]. Studying the effects of desexing on lifespan also adds some confounders in that at least in the Western context, desexing is often a proxy for overall better veterinary care, which may result in an overestimation of any beneficial effect of desexing on lifespan. In addition, any desexed dog must necessarily already have lived to the age at which it was desexed, which skews the desexed population towards older dogs by removing any dog that died before the age at which desexing would have been carried out [107]. Furthermore, the presence of right censored data may further skew the results of lifespan studies based exclusively on death records by resulting in underestimated lifespans that may not be consistent between the groups considered [109].

The first study including systematic information on lifespan and desexing was published in 1982 based on necropsy data from n = 2002 dogs from a private veterinary hospital in New England covering necropsies performed between 1962 and 1976 [110]. This study found a significant lifespan advantage of gonadectomized over intact dogs in both sexes, with intact females living to a mean age of 4.7 years, while gonadectomized females lived to a mean age of 8.6 years. In the same study, intact males lived to a mean age of 4.8 years, while gonadectomized males lived to a mean age of 9.9 years. When considering only dogs that had lived to at least 2 years of age, this was 7.7 vs. 8.8 years in intact vs. gonadectomized females, and 8.0 vs. 9.9 years in intact vs. gonadectomized males. Interestingly, this study did not find a body size effect on lifespan, which contradicts much of the subsequent literature on body size and lifespan in dogs.

Another study investigated lifespan in dogs in the UK based on owner surveys representing n = 3121 deceased dogs, and found that intact females had died at a median age of 12 years, while gonadectomized females had died at a median age of 10 years, 10 months. In the same study, intact males had died at a median age of 10 years, 11 months, while gonadectomized males had died at a median age of 10 years, 8 months. The authors reported that desexed females lived significantly longer than intact females as well as males regardless of desexing; however, there was no significant difference in lifespan between desexed and intact males [103].

In 2013, a study based on the Veterinary Medical Database (VMDB), which aggregates data on deceased dogs from US veterinary teaching hospitals, found that based on data covering the years 1984–2004 from n = 70,574 deceased dogs, desexing increased male life expectancy by 13.8%, while it increased female life expectancy by 26.3% [108]. Additionally, in a later analysis based on the same VMDB data and this time including juvenile dogs and dogs of unknown desexing status (n = 80,958), it was found that intact males lived significantly longer that intact females, but gonadectomized females were the longest living demographic. The same study also analyzed data from the VetCompass database of private veterinary practices in the UK spanning the time period 2009–2011 (n = 5095 deceased dogs) and came to the same conclusion [105].

A breed-specific study based on n = 652 deceased Golden Retrievers with known ages at death necropsied at a US veterinary teaching hospital between 1989 and 2016 found that gonadectomized females lived significantly longer than intact females; however, there was no significant lifespan advantage of gonadectomized over intact males [111]. In contrast, an international study based on an online survey of n = 2505 Vizslas born between 1992 and 2008 found no difference in lifespan between intact and gonadectomized dogs [112].

Most recently, data based on the Banfield network of primary care veterinary clinics in the contiguous US including ages at most recent visit in addition to known death data (n = 2,504,518) found that intact females lived to a median age of 13.77 years; gonadectomized females, 14.35 years; intact males, 14.09 years; and gonadectomized males, 14.15 years [107].

In summary, all large-scale studies of the effects of desexing on lifespan are based on data from either the US or the UK, all of them investigate gonadectomy as opposed to other methods of desexing,

all but one exclusively rely on data from deceased dogs, and none of them take age at desexing into account. Additionally, the available studies consistently find a survival advantage in gonadectomized over intact females, while the data in males are inconsistent, and where such an effect is described in males, it is consistently of lesser magnitude than is the case in females. This is potentially problematic when considering desexing across both sexes as part of a study, as the stronger effect in females may overshadow the weaker or non-existent effect in males [107].

The issue of age at desexing in lifespan studies is an important question that has not been addressed in any large-scale studies thus far. In this context, it is likely that the amount of time during which a dog is exposed to sex hormones influences its disease risks and overall life course [112,113].

5.2. Individual Disease Risks

5.2.1. Existing Review Articles

A large body of literature exists covering various long-term health aspects of desexing, with most papers focusing on individual disease and/or cause of death risks; however, many of the existing studies on individual disease risks are limited to one breed or to a relatively low number of breeds [104,112,114–120].

Various general review papers giving overviews of the reported disease risk changes associated with desexing or not desexing have been published at this point [15,78,121,122], including several reviews specifically focusing on the long-term health effects of pediatric desexing [16,123–126]. In addition, several disease-specific review articles exist examining the influence of desexing on cancer development [127], mammary tumor risk [128], pyometra [129], and urinary incontinence in female dogs [130].

As explained above, our view is that individual disease risks are considerably less important than lifespan when measuring health outcomes of desexing in dogs, and given the extensive amount of existing review articles including articles on individual disease risks, we have aimed to keep this part of the present review fairly concise. Based on the extent of the available disease-specific literature, we will focus on diseases of the reproductive system *sensu lato*, tumors not involving the reproductive system, joint disorders, urinary tract disease, and weight gain and obesity.

5.2.2. Diseases of the Reproductive System

Seeing as gonadectomy involves removal of the gonads, it eliminates the risk of any diseases directly affecting these organs. In males, this includes the various forms of testicular tumors, which consist of Leydig cell tumors, Sertoli cell tumors, interstitial cell tumors, seminomas, and teratomas [131–140]. Testicular tumors predominantly occur in aged dogs, and more than half of all testicular tumors occur in cryptorchid dogs [131]. A breed predisposition has been described in Maltese, which may be linked to an increased risk of cryptorchism in the breed [131]. In females, the ovarian tumors described in the literature include epithelial tumors, granulosa cell tumors, Sertoli–Leydig cell neoplasms, sex cord stromal tumors, and germ cell tumors including teratomas [141–145].

In females, ovariohysterectomy includes the removal of the uterus, eliminating the risk of various uterine diseases, including pyometra. In a study of over 200,000 insured dogs in Sweden, the 12-month pyometra risk in intact female dogs aged under 10 years was 2%. This study found breed predispositions in Rough Collies, Rottweilers, Cavalier King Charles Spaniels, Golden Retrievers, Bernese Mountain Dogs, and English Cocker Spaniels, while breeds with below-average risk were Drevers, German Shepherds, Dachshunds, and Swedish Hounds [146]. A similar study on 260,000 insured Swedish dogs found that the mean age of diagnosis was 8 years, and that 19% of all intact females had been affected with pyometra by age 10 [6].

Gonadectomy in female dogs is commonly thought to result in a significant reduction in mammary tumor risk, especially if performed prior to the first estrous cycle. Mammary tumors are some of the most commonly diagnosed tumors in intact female dogs, with roughly 90% of them being epithelial

in nature, with the remaining 10% being sarcomas and mixed type tumors [147]. Roughly 50% of mammary tumors in female dogs are benign, and 50% of the malignant tumors lead to metastasis [148]. In the largest study thus far, 13% of 260,000 insured intact female dogs had developed mammary tumors by age 10 [6].

There are multiple studies supporting the view that gonadectomy (particularly when done early in life) has a protective effect against mammary tumors in female dogs [120,149–151]; however, a recent review of the literature based on Cochrane guidelines judged that the existing evidence in favor of this view is weak overall [128].

In males, desexing eliminates the risk of benign prostatic hyperplasia, which can be found in more than 95% of aged intact male dogs. However, it only causes clinical signs in a relatively small subset of these dogs, and curative treatments are available medically or by desexing [152–156]. Desexing also decreases the risk of perianal adenomas in male dogs [157]. Surprisingly, desexing increases the risk of malignant prostate tumors in dogs, with gonadectomized dogs having 3.9 times the risk of intact ones in a study on 70 dogs. Most of these prostatic neoplasms in dogs were carcinomas of ductal origin and were not androgen-dependent [158]. However, this is a relatively rare disease, which limits its importance in the decision making process in male dogs [122].

Finally, desexing has been found to decrease the prevalence of transmissible venereal tumor across both sexes in areas where the disease is enzootic, presumably by reducing the number of sexual encounters in the population associated with the loss of libido in desexed dogs [159].

In summary, the evidence suggests a beneficial effect of desexing on the occurrence of common and commonly life-limiting diseases of the reproductive system in female dogs, particularly concerning pyometra and (to a somewhat lesser degree) mammary tumors, with large variance in breed-specific pyometra risks. In males, the beneficial effects of desexing are most dramatic on benign prostatic hyperplasia, which is a common, but not generally life-limiting disease, while the influence of male desexing on life-limiting diseases is mixed and limited to comparably rare diseases.

5.2.3. Tumors not Involving the Reproductive System

Desexing has been shown to increase the risk of several non-reproductive tumors in dogs. These risk increases can affect both sexes or may only be present in females, depending on the tumor. In contrast, apart from perianal adenomas in male dogs mentioned under the previous point, there are no non-reproductive tumors that show a risk decrease following desexing in dogs, and no non-reproductive tumor risk increases that are specific to desexed male dogs [127].

Non-reproductive tumors with an increased risk across both sexes following desexing are cardiac and splenic hemangiosarcoma [112,117,160–162], appendicular osteosarcoma [162–165], lymphoma [112,117,162,166], and transitional cell tumors of the bladder [167,168]. An increased risk for mast cell tumors has been described that may be specific to female dogs [120,162].

There have been a few breed-specific studies on gonadectomy and tumor risk. In a study on Golden Retrievers based on dogs necropsied at a US veterinary teaching hospital between 1989 and 2016, it was found that a greater proportion of gonadectomized females died of cancer than was the case in intact females; however, gonadectomized females also lived longer, and unsurprisingly, age was found to be a more important factor for cancer risk than gonadectomy. In the same study, gonadectomized male dogs did not live longer than intact males, and there was no difference in tumor related mortality [111]. A study based on patient records of n = 1015 Golden and n = 1500 Labrador Retrievers from the same US veterinary teaching hospital between 2000 and 2012 found that the increase in tumor risk following gonadectomy was strongest in female Golden Retrievers, and that the tumor risk increases associated with gonadectomy were stronger in this breed as compared to Labrador Retrievers [114]. Additionally, a study of n = 1170 German Shepherds from the patient records of the same veterinary teaching hospital collected between 2000 and 2014 found no difference in cancer risk in gonadectomized over intact males, and with the exception of mammary tumors, there was no increased cancer risk in gonadectomized over intact females [120].

In a cohort of n = 2505 Vizslas born between 1992 and 2008 that was studied through a worldwide online survey, it was found that gonadectomy increased tumor risk across both sexes. In this study, mast cell tumors were diagnosed at significantly earlier ages in gonadectomized as opposed to intact dogs, while there was no difference in ages at diagnosis in lymphoma and hemangiosarcoma. Other cancers were also diagnosed at significantly earlier ages in gonadectomized dogs as compared to intact ones across both sexes. This study found no difference in lifespan between intact and gonadectomized dogs overall [112].

In summary, there is consistent evidence for an increased osteosarcoma, lymphoma, transitional cell cancer, and hemangiosarcoma risk in gonadectomized dogs of both sexes, while the evidence is inconsistent for other types of tumors and suggests that there may be considerable variation between breeds and sexes. Additionally, and as mentioned above, the most important risk factor for tumor development is age [99–101], and many of the present studies on tumor risk do not take age into account, meaning that some of these findings may also reflect a difference in lifespan as opposed to a direct influence of desexing on tumor development. Unsurprisingly, when studies take age into account, it is shown to be a more important tumor risk factor than gonadectomy [111]. In addition, many studies are limited to one or a relatively low number of dog breeds, which may represent breed-specific risk changes and therefore raise concerns regarding generalizations to the whole dog population.

5.2.4. Joint Disorders

Desexing has been shown to affect the risk for various joint disorders, both across the general dog population and in individual breeds. Disorders that have been reported as being more common in desexed dogs include craniate cruciate ligament (CCL) injury, hip dysplasia (HD), elbow dysplasia (ED), and osteoarthritis in general. The most consistently reported joint disorder with an increased risk in gonadectomized dogs of both sexes is CCL injury, which has been reported both in the general population and in individual breeds such as German Shepherds as well as Golden and Labrador Retrievers [114,116,117,120,162,169,170], and with some of the available studies reporting an increased risk in dogs that were desexed before puberty.

Other joint disorders that have been found to occur more commonly in desexed dogs are HD in gonadectomized male Golden Retrievers [117]; overall joint disorder risk (HD, ED, CCL rupture) in gonadectomized German Shepherds of both sexes [120], Golden Retrievers of both sexes [114], and female Labrador Retrievers [114]; ED and CCL rupture in male Labrador Retrievers that were gonadectomized before six months of age [114]; and CCL injury and osteoarthritis in Golden Retrievers of both sexes that were gonadectomized before six months of age [116].

While one study has described a decrease in bone density in ovariectomized laboratory Beagles [171], this result has not been reproduced in a later study [172], and to the authors' knowledge, the veterinary literature does not contain any indications that ovariectomy is a risk factor for clinically relevant bone density loss in female dogs.

In summary, the risk of CCL injury appears to consistently increase with desexing across the general dog population as well as in the individual breeds studied, while there appears to be considerable variability in HD, ED and osteoarthritis risk increase depending on breed, sex, and age at desexing. More research is needed particularly in the fields of breed- or size-specific ED and HD risk, as well as age at desexing.

5.2.5. Urinary Disease

There is evidence from multiple studies that gonadectomy increases the risk of urinary incontinence in female dogs, and that this risk increase is also associated with larger body size [120,125,173–178], with a prevalence of urinary incontinence of up to 20% of gonadectomized females reported in the literature [179]. This appears to be a sex-specific effect, as orchiectomy does not appear to be associated with an increased incontinence risk in male dogs [180]. However, a 2012 systematic review of the literature based on Cochrane criteria concluded that the evidence for an increased risk of urinary

incontinence in gonadectomized female dogs was weak, and most consistent in female dogs that had undergone the procedure before three months of age [130]. Apart from urinary incontinence, gonadectomized female, but not male dogs have also been reported to be at a higher risk of struvite urolithiasis as compared to intact females and intact males respectively [181].

Interestingly, it has been reported that urinary incontinence in gonadectomized female dogs can be successfully treated using GnRH superagonist implants, which as we report under point 3.2 are an increasingly common means of desexing in Western Europe and Australia [179,182]. These implants work by suppressing the release of gonadotropins, while surgical gonadectomy increases the release of gonadotropins due to the removal of the negative feedback mechanism provided by the presence of sex steroids secreted in the ovaries. This in turn would indicate that the urinary incontinence risk increase seen in gonadectomized female dogs is not caused by the lack of sex steroids associated with desexing, but rather by the increased levels of gonadotropins induced by ovariectomy.

In summary, the effects of desexing on urinary incontinence and urolithiasis risk increase appear to be limited to female dogs. There is a substantial amount of evidence for ovariectomy and ovariohysterectomy being associated with an increased risk of urinary incontinence, particularly in large dogs that have been desexed at an early age. This effect may be specific to desexing methods involving ovariectomy and may not be associated with methods that do not affect the release of gonadotropins.

5.2.6. Weight Gain and Obesity

There are multiple studies from a wide variety of geographic areas linking desexing to an increased risk of obesity in dogs. These studies include dog populations from the US, the UK, Continental Europe, Australia, China, and Japan [116,183–191]. While one recent study performed in Danish dogs found an increased obesity risk associated with desexing that was limited to male dogs [189], most other available studies find that the effect is present in both sexes. In addition, an international study of 926 overweight dogs undergoing a weight loss diet for three months found that intact dogs of both sexes lost significantly more weight during the study period than desexed dogs [192].

One US study comparing patient populations between a low-cost and a general practice veterinary clinic found that obesity in dogs was not associated with socioeconomic status of the owners, but that desexed dogs were consistently more likely to be obese across both populations [183]. In addition, diabetes mellitus has been reported to be more common in neutered male dogs as compared to intact males, and obesity is a known risk factor for diabetes mellitus in dogs [193]. Interestingly, in one study that measured food intake in free-fed intact female dogs across their sexual cycle, it was found that caloric intake was lowest during estrus and highest during anestrus, and that caloric intake increased after gonadectomy, indicating that the presence of female sex steroids may have a negative effect on caloric intake [194].

In summary, there is consistent evidence that desexing is associated with an increased risk of obesity in dogs of both sexes across a wide variety of environments, as well as some evidence that weight loss is more difficult to achieve in desexed as compared to intact dogs, and that caloric intake is decreased by the presence of sex steroids at least in female dogs.

5.2.7. Other Disease Risks

Various other disease risk increases associated with desexing in one or both sexes have been published for both the overall dog population and in studies limited to specific breeds:

A retrospective study based on n = 90,090 patient records from a US veterinary teaching hospital found that gonadectomized dogs of both sexes were at a significantly higher risk of immune-mediated diseases including atopic dermatitis, autoimmune hemolytic anemia, Addison's disease, hypothyroidism, immune-mediated thrombocytopenia, and inflammatory bowel disease, with the effect being stronger in females compared to males except for autoimmune hemolytic anemia and Addison's disease [195]. In this context, a study on Golden and Labrador Retrievers found that

desexed dogs had lower IgE levels than intact dogs in both sexes; however, this difference was not associated with clinical disease [115].

Another retrospective study based on the same veterinary teaching hospital patient records found that gonadectomized dogs of both sexes were less likely to be diagnosed with several congenital or early-onset inherited disorders including aortic stenosis, early onset cataracts, patent ductus arteriosus, ventricular septal defect, and portosystemic shunt. The same study also found that gonadectomized males were less likely to be diagnosed with gastric dilation volvulus and dilated cardiomyopathy as compared to intact males, and that gonadectomized dogs of both sexes had an increased risk of hyperadrenocorticism and epilepsy [162]. In this context, it is likely that the reduced risk of congenital or early-onset inherited disease reflects a selection bias within the gonadectomized population as opposed to an increased baseline risk.

As mentioned under the previous point, diabetes mellitus has been reported to be more common in desexed male, but not female dogs compared to intact dogs across the whole dog population [193].

Finally, one study of dogs affected with idiopathic epilepsy based on the VetCompass database using Labrador Retriever and Border Collie patient records from private veterinary practices in the UK found that 74% of gonadectomized patients were gonadectomized before the onset of seizures, and that dogs that had been gonadectomized before the onset of seizures had significantly longer survival times than those that had been gonadectomized afterwards [119].

In summary, a wide variety of additional disease risk increases and decreases has been reported in the literature; however, most of these findings have not thus far been independently reproduced.

6. Behavior

The desexing methods that result in the elimination of sex steroids have been shown to affect behavior in multiple studies. As is the case in the literature on other aspects of desexing, most studies are specific to surgical gonadectomy; however, there are also a few studies that have examined the effects of non-surgical methods, most prominently those of GnRH superagonist implants and progestagen administration.

Review articles exist on the topic of desexing effects on canine behavior for both the effects of gonadectomy [5,196] and GnRH superagonist treatment [197]. Most studies on behavioral changes associated with desexing published thus far have been specific to male dogs, while comparably few of them have also included or focused on behavior in gonadectomized female dogs. In studies that differentiated between effects in males and females, the behavioral effects of desexing were generally found to be more pronounced in males than they were in females [190].

An important clinical issue in desexing and behavior is presented by the difficulty to predict which desirable or undesirable behavioral changes will occur in which individual dogs [198], which is particularly concerning given the irreversible nature of gonadectomy. In this context, we note that in one study based on companion dog owner surveys in Australia, 61% of male dog owners and 47% of female dog owners stated that they would not proceed with desexing their dog if they were to be given the same choice again [199]. Where available, fully reversible desexing methods such as GnRH superagonist implants are therefore a particularly valuable tool in situations where desexing is considered as a behavioral intervention, as treatment can be discontinued and the hormonal *status quo ante* restored in case of undesirable outcomes [197].

Based on the existing literature, the effects of desexing on behavior can be divided into libido and its associated behaviors such as roaming and certain aspects of mounting, urinary marking, dog bite injury risk in humans and conspecifics, various other forms of boldness-related, aggressive or reactive behaviors, and cognitive function.

6.1. Libido and Associated Behaviors

Testosterone controls libido and attraction to estrous females, which first appear in intact male dogs between ages 4 and 6 months [200]. Sexual mounting and copulatory behavior is mediated by

testosterone effects on the medial preoptic-anterior hypothalamus [201]. While desexing of sexually inexperienced dogs generally eliminates libido and copulatory behavior, attraction to female dogs including copulatory behavior may persist in sexually experienced male dogs even after desexing [202]. GnRH superagonist implants initially stimulate the pituitary-gonadal axis, which may result in a transient increase in libido and associated behaviors following the initiation of treatment [197].

Several studies have found that gonadectomy significantly reduces mounting of conspecifics and roaming behavior in 60% to 90% of male companion dogs [191,203,204]. However, one study found increased mounting of objects in desexed males [205], and in a study on free-roaming dogs, neither roaming nor mounting behavior was reduced in free-roaming dogs undergoing either surgical or hormonal desexing when compared to their intact conspecifics [206]. Various studies have found no effect of age at desexing on its effects on either roaming or mounting behavior [5,198,203]. In this context, it is important to note that mounting behavior can also occur for reasons not related to libido, such as stress or anxiety, and that this would most likely contribute to the observed lack of effect of desexing on mounting behavior in some cases [207].

In summary, and unsurprisingly, the literature consistently shows that desexing reduces libido and its associated behaviors such as conspecific mounting and roaming in the majority of desexed male companion dogs. Sexual behavior persistence after desexing appears to depend on previous sexual experience rather than age at desexing. Interestingly, desexing effects on conspecific mounting and roaming behaviors do not appear to be present in free-roaming dogs.

6.2. Urinary Marking

Inappropriate urinary marking in male companion dogs is a behavior that may cause considerable distress to their owners and accounts for a significant percentage of dogs being relinquished to shelters [208,209]. Male-pattern urinary marking is a testosterone-dependent behavior that is initiated during puberty [210,211], but which contrary to mounting and copulatory behavior does not depend on testosterone effects in the preoptic-anterior hypothalamus [201]. The majority of studies of desexing found a significant decrease in urinary marking behavior in desexed male dogs regardless of age at desexing [5,190,191,203,204], while one study found an influence of age at desexing, with later ages being associated with a less pronounced decrease in marking following desexing [205]. In contrast, female dogs do not generally show urinary marking, and urinary behavior is not generally affected by desexing in female dogs [212].

In summary, there is consistent evidence across studies that desexing reduces the frequency of urinary marking in the majority of male dogs, and mostly consistent evidence that this effect does not depend on the age at desexing.

6.3. Bite Injury Risk

Dog bite injury risk is an aspect of dog behavior that has substantial societal and public health implications. Dog bites cause an estimated 580,000 human injuries and an average of 20 human deaths in the US per year, with 51% of reported dog bite cases occurring in children under age 12 [213]. Consequently, a considerable body of literature based on dog bite injury report data exists, some of which also includes the effects of desexing [214–217].

A systematic literature review based on observational studies of dog bite risk concluded that five out of six of the articles considered showed that intact dogs were more likely than desexed dogs to cause bite injuries; however, the authors also concluded that the available data was insufficient to estimate the effect size, and that all studies considered were observational case-control studies rather than interventional studies [196]. All of these studies were based on dogs from the US and/or Canada, and in this context, as is the case for health outcomes, it should be considered that desexing may act as a proxy for overall better animal husbandry standards in these populations.

Dog bite risk has also been studied in free-roaming dog populations. In one such study, a street dog desexing program in Jaipur, India resulted in a decrease of human bite injuries that the authors attributed to a decrease in female dogs showing maternal protection behavior [96].

In summary, there is mostly consistent evidence for a reduction in human bite injury risk associated with desexed dogs; however, in the companion dog population this is confounded by the fact that all studies thus far have been performed in populations where desexing acts as a proxy for overall better animal husbandry. In practice, a thorough behavioral evaluation of any dog with a bite history is paramount to identify the underlying motivation for the bite incident [218].

6.4. Boldness-Related and Aggressive Behavior

The effect of desexing on boldness-related and aggressive behavior has also been considered in a variety of studies based on owner questionnaires and surveys using both the overall dog population and specific breeds. Boldness is defined as one end of the shyness–boldness axis in dog personality that incorporates several different personality traits such as playfulness, sociability including fear of or aggression towards people, and curiosity/fearlessness [219–221]. It can be reproducibly measured in standardized owner questionnaires such as C-BARQ [222]. Aggression is a trait within the boldness spectrum that has been fairly extensively studied in dogs, and is generally divided into aggression directed at humans and aggression directed at other dogs.

One study on n = 1054 companion dogs in Australia using the C-BARQ questionnaire found that male dogs scored higher in boldness traits than female dogs, and that desexed dogs of both sexes scored significantly lower in boldness traits than intact dogs [223]. On the other end of the shyness-boldness spectrum, several studies have reported an increase in fearfulness and/or reactivity associated with gonadectomy. This included increased fearfulness in gonadectomized female Labrador Retrievers [212] and gonadectomized Vizslas of both sexes [112], as well as reactivity and vocalization in ovariectomized German Shepherd females [224,225].

A study of n = 13,796 dog owners using the C-BARQ questionnaire found no effect of gonadectomy on aggression towards familiar people or dogs across both sexes, but found a small but significant increase in aggression towards strangers in dogs that had been gonadectomized between ages 7 and 12 months [226]. A different study also using the C-BARQ questionnaire and focusing on n = 6235 male dogs that had been orchiectomized for non-behavioral reasons before age 10 years found that a longer percentage of lifespan spent desexed was associated with both increased fearfulness and increased aggression [205].

Two studies based on owner questionnaires of n = 2207 dogs found that desexed males were significantly more likely to show both dog-directed and human-directed resource guarding aggression than all other demographics [227,228].

Several studies have investigated various forms of aggression in male dogs that were orchiectomized for behavioral reasons using owner interviews or questionnaires. One such study of 57 dogs found that aggression towards family members was reduced in 30% of dogs, while aggression towards strangers or intruders was reduced in 10% of dogs following orchiectomy [198]. In contrast, similar a study of n = 122 orchiectomized male dogs found that inter-male dog aggression was reduced in about 60% of dogs following orchiectomy [191]. A similar retrospective study of n = 42 orchiectomized male dogs found reduced aggression towards other male dogs in 60% of dogs, but no reduction in territorial or fear-induced aggression [203].

Finally, an observational study on n = 93 Hungarian dogs that measured both obedience and owner-directed aggression found that males and intact dogs tended to be less obedient than females and gonadectomized dogs, but that gonadectomized dogs that were less obedient were also more likely to show signs of owner-directed aggression [229].

In summary, the evidence for an influence of desexing on boldness-related and aggressive behavior is inconsistent and sometimes contradictory. The most consistent finding appears to be a reduction in dog-directed aggression in male dogs that have been orchiectomized due to behavioral problems;

however, this conclusion is based on a comparably low number of dogs and may not apply to males without pre-existing behavioral issues. Similarly, in studies that did not include information on the owners' motivation for desexing, there may have been a selection of dogs that were showing problematic behaviors before desexing, which also raises issues regarding the potential of misdiagnosis in case a thorough behavioral assessment was not carried out. In addition, many of the available studies do not differentiate between males and females when considering behavioral outcomes of desexing; however, considering the important differences in endocrine physiology between the sexes, it would not be surprising to find that effects of desexing on boldness-related and aggressive behavior could differ between the sexes.

6.5. Cognitive Function

Some literature exists on the impact of gonadectomy on cognitive function in dogs, most notably regarding Canine Cognitive Dysfunction (CCD), which is a behavioral syndrome affecting older dogs that shares many pathophysiological and behavioral hallmarks with human Alzheimer's Disease (AD), including progressive cognitive impairment, loss of normal sleep patterns, increased anxiety, and aimless wandering as well as Amyloid-beta and possibly tau pathology in the brain [230–232]. In this context, female and gonadectomized dogs have been shown to be significantly more likely to show signs of CCD than intact and male dogs in a study of n = 325 geriatric dogs aged nine or older [233]. CCD has also been shown to progress more rapidly in gonadectomized male dogs that already have signs of mild cognitive impairment compared to intact males with mild cognitive impairment during a 12–18 month follow-up period [234]. Interestingly, this mirrors findings in human patients, where women have an increased AD risk as compared to men [235], and where men receiving androgen deprivation therapy for prostate cancer have been shown to be at a higher risk of developing dementia and AD than controls [236].

Apart from CCD, there is some limited evidence from standardized behavioral tests on companion dogs that found cognitive differences related to desexing. In one such study on n = 64 dogs that were grouped by sex and gonadectomy status and tested in a spatial learning maze task, the authors found that intact females had the highest success rates of all four groups in both learning and completing the task, as well as memorizing it after a latency period of two weeks, while there was no difference between ovariectomized females, intact and orchiectomized males in either outcome [237]. Interestingly, female, but not male dogs have also been shown to react to a size consistency violation in an object display test using 25 male and 25 female companion dogs; however, there was no difference between orchiectomized and intact males [238].

In summary, there is some limited evidence that gonadectomy increases the risk of cognitive dysfunction in both sexes, and increases the speed of progression from mild to more severe cognitive impairment in male dogs, which interestingly seems to mirror sex-specific aspects of Alzheimer's Disease in humans. There is also evidence for differences in spatial learning between male and female dogs that appear to be influenced by ovariectomy, but not orchiectomy; however, the field is very much in its infancy and more research is needed to further characterize these differences and how they are influenced by desexing.

7. Conclusions

While a wide variety of surgical and non-surgical desexing methods have been described in the literature, in practice gonadectomy is by far the most commonly used desexing method, and the great majority of studies on desexing have been limited to gonadectomy as opposed to other forms of desexing. Desexing rates are highly variable depending on geographic area even within the Western world. The terminology used to describe various desexing methods is inconsistent and contradictory, and there appears to be a need for the adaption of standardized terminology in the literature.

is the available empirical evidence shows a clear lack of an effect of desexing as a population control measure in companion and shelter dogs, which is surprising given that desexing is commonly

recommended in the literature to achieve this goal. However, there is evidence that desexing is effective for population control in free-roaming dog populations and may have unrelated health benefits in these populations by reducing bite risk and risk of zoonotic disease transmission due to combined desexing and vaccination efforts. Mathematical modelling studies indicate that most of the population control effect in these populations can be attributed to female rather than male desexing.

A large amount of literature related to disease risk increases and decreases associated with desexing exists, but commonly neglects to consider the effect of age as the most important risk factor for most of the studied diseases. The largest effects on common and commonly life-limiting diseases appear to be seen in the female reproductive tract, most notably with regards to pyometra and, to a lesser extent, mammary tumors. While some disease risks appear to be consistently affected by desexing across both sexes and all breeds, other risk changes appear to be sex- and/or breed-specific, and more research is needed to determine the extent of variation between breeds and sexes for these diseases. Disease risks that are consistently reported to be increased with desexing in both sexes and across breeds include cranial crucial ligament rupture, hemangiosarcoma, lymphoma, transitional cell carcinoma of the bladder, and obesity.

In contrast, there are relatively few studies on the effects of desexing on lifespan in dogs, and the available studies are subject to multiple biases associated with desexing being a proxy for overall better husbandry and veterinary care, desexed dogs already having lived to the age at which the procedure was performed, and right censored data in case of studies based on death records. However, when taken together, there is consistent evidence for a beneficial effect of desexing on lifespan in female dogs, and inconsistent evidence for such an effect in male dogs, and the male-specific effect is consistently shown to be of lower magnitude than the female-specific effect if present.

The amount of literature on the behavioral outcomes associated with desexing is larger for male than for female dogs and consistently shows a reduction in libido, roaming, conspecific mounting, and urinary marking frequency in a large percentage of orchiectomized males, as well as a reduction in inter-male dog aggression in male dogs that were orchiectomized because of existing behavioral problems. Limited evidence also exists for desexing increasing the risk and progression of age-related cognitive dysfunction. In addition, there is some evidence that desexed dogs of both sexes are less likely to cause bite injuries to humans; however, these studies have been conducted in environments where desexing is expected to act as a proxy for overall better husbandry.

The evidence for other behavioral effects such as human-directed aggression, human or object mounting, resource guarding, shyness and anxiety-related behaviors is inconsistent and sometimes contradictory, and often fails to differentiate between the sexes. There are relatively few studies specific to females or individual breeds, and more research is needed to further elucidate the influence of desexing in these populations.

One additional limitation of the available literature on the various effects of desexing is that most of it reflects post hoc observational studies, not randomized interventional studies. As such, most of the effects described in the literature are necessarily correlations that do not strictly establish causality with desexing. Where feasible, more research is needed using prospective study designs.

In conclusion, we find that the evidence for an overall beneficial effect of desexing is stronger in female than in male dogs when considering both population control and health benefit aspects; however, the available evidence suggests that there is significant variation between the breeds and sexes, and more research is needed to further elucidate these differences and to arrive at more individualized evidence-based recommendations for clinical practice.

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